MULTIMODAL AI FRAMEWORK FOR EARLY MULTI-DISORDER NEURODEVELOPMENTAL SCREENING AND RISK ASSESSMENT

A CAPSTONE PROJECT REPORT

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DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted elsewhere, partially or fully, for the award of any other degree or diploma. Relevant previous work presented in this report has been properly acknowledged and cited. The plagiarism policy, as stated by the project guide and project coordinator, has been maintained.

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ABSTRACT

NeuroScan AI is a multimodal artificial intelligence framework proposed for early screening and risk assessment of neurodevelopmental and neuromental disorders in children, particularly in underserved communities where access to specialist assessment may be limited. The framework is intended to combine relevant modalities such as audio, video, behavioural observations, and structured input, where available, to identify patterns that may indicate the need for further professional evaluation. The proposed workflow includes data collection with appropriate consent, preprocessing, feature extraction, multimodal analysis, and presentation of a screening-oriented risk indication through a web-based interface. The system is designed as a decision-support and early-screening aid rather than a replacement for diagnosis by qualified healthcare professionals. Particular attention is given to privacy, responsible data handling, accessibility, and the limitations of datasets and model generalisation. The expected outcome is a prototype that can help users recognise potential developmental concerns earlier and encourage timely referral for professional assessment.

The project aims to support accessible early screening through a web-based multimodal AI workflow while maintaining privacy, transparency, and appropriate clinical boundaries.

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LIST OF ABBREVIATIONS

ACRONYM	DEFINITION
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
ML	Machine Learning
NDD	Neurodevelopmental Disorder
UI	User Interface

CHAPTER 1 INTRODUCTION

1.1 Background and Motivation

Neurodevelopmental disorders can affect communication, learning, behaviour, and everyday functioning. Early identification of developmental concerns can help children and families access appropriate assessment, support, and intervention. However, specialist services may be difficult to access in underserved communities because of limited awareness, cost, travel distance, and shortages of trained professionals. Conventional screening often relies on observation, interviews, and questionnaires, which remain important but may require trained personnel and repeated visits. Recent advances in artificial intelligence make it possible to analyse patterns across different data types, including audio, video, and structured behavioural information. NeuroScan AI explores a multimodal framework that combines available modalities to support preliminary screening and risk assessment. The motivation is to develop an accessible prototype that can help flag possible concerns and encourage timely professional evaluation without claiming to provide a definitive diagnosis.

1.2 Problem Statement

Children who may show early signs of neurodevelopmental or neuromental disorders can face delays in receiving an appropriate assessment, especially in underserved areas. Existing screening pathways may be limited by access to specialists, the time required for manual assessment, and the fact that a single data source may not capture a child's behaviour comprehensively. The project addresses the need for an accessible, web-based screening support framework that can analyse multiple available modalities and present a preliminary risk indication. The output is intended to support awareness and referral, not to replace clinical judgement or professional diagnosis.

1.3 Objectives and Outcomes

The main objective is to design a multimodal AI framework for preliminary screening and risk assessment of neurodevelopmental and neuromental disorders in children. The specific objectives are to: (i) identify suitable audio, video, and structured behavioural input modalities; (ii) define preprocessing and feature-extraction steps for each available modality; (iii) investigate a model architecture that combines modality-specific information; (iv) provide a simple web interface for submitting inputs and viewing screening-oriented results; (v) consider privacy, consent, accessibility, and bias during

system design; and (vi) define a testing approach using appropriate datasets and evaluation metrics. The intended outcome is a prototype workflow that demonstrates how multimodal analysis may support earlier identification of concerns.

1.4 Scope and Applications

The project covers the design of a web-based prototype for multimodal early screening and risk assessment. Depending on dataset availability and permissions, the framework may process audio, video-derived behavioural features, and structured inputs through separate preprocessing stages before combining the available features for model inference. The interface is intended to communicate a preliminary risk indication and encourage consultation with qualified professionals when concerns are identified. The project does not provide a medical diagnosis, prescribe treatment, or replace developmental assessment by clinicians. Potential applications include awareness and referral support in community programmes, educational settings, and underserved areas, subject to validation, consent, privacy safeguards, and professional oversight. The framework is intended for research and prototype evaluation only; real-world clinical use would require further validation and appropriate app.

CHAPTER 2 STATE OF ART AND NEED

2.1 Existing Research, Technologies or Products

Present a comprehensive review of the state of the art relevant to the project. Use proper IEEE citations for research articles and patents. Include comparative tables or figures where useful to contrast multiple approaches. The goal is to demonstrate that the team understands the current landscape before proposing their own solution [1].

2.1.1 Research Literature

This is compulsory. Summarize key academic studies, methods, and findings in the field (e.g., AI-based depression prediction models, clinical screening methods) [2].

2.1.2 Existing Technologies or Commercial Products

This is compulsory. Discuss currently available tools, devices, or software solutions, out-lining their features, costs, and limitations.

2.1.3 Relevant Patents

This is optional. Highlight any patented technologies that show how the problem is currently being addressed in industry and identify opportunities to innovate without infringement [3].

2.2 Gaps and Challenges in Existing Approaches

[Discuss the **weaknesses, shortcomings, or challenges** in current methods. For exam-ple: existing depression screening methods are subjective, costly, or lack real-time capa-bility. Many AI-based studies rely on limited datasets or fail to consider diverse popula-tions. This section should lead logically to why a **new solution is needed**. Use evidence and **cite sources in IEEE format**. [4]]

2.3 Need for the Proposed Project

[Summarize **why this project is necessary**, based on the gaps identified in the previous section. For an *automatic depression prediction project*, students can explain that existing diagnostic methods are limited (e.g., time-consuming, subjective, or costly), while exist-ing AI models often lack real-world validation, scalability, or accessibility. They should highlight how these shortcomings create a **pressing need** for a new solution. This section **should not describe the final design (that comes in Chapter 4) but should establish the justification for undertaking the project itself**. Where possible, use supporting statistics, references, or standards in IEEE format to strengthen the argument. [5]]

2.1 Existing Research, Technologies or Products

Early screening of neurodevelopmental and neuromental disorders in children is an important application of artificial intelligence. Traditional screening methods involve clinical observation, developmental assessments, questionnaires, and specialist consultations. Research in artificial intelligence has explored machine learning and deep learning techniques to identify patterns in behavioural, speech, facial, and other relevant data. Multimodal approaches combine information from multiple sources and may provide a more comprehensive assessment than relying on a single data type. Existing digital screening tools can improve accessibility, but their performance depends on data quality, model reliability, and appropriate clinical validation. Commercial and research systems vary in features, availability, and cost. Their limitations include restricted datasets, limited demographic diversity, privacy concerns, and the need for professional interpretation. Relevant patents in AI-based health assessment also demonstrate ongoing innovation in automated screening and decision-support technologies.

2.2 Gaps and Challenges in Existing Approaches

Existing approaches face several challenges. Conventional assessments may require trained specialists, repeated consultations, and considerable time, limiting access in underserved communities. Single-modal AI systems may overlook important behavioural indicators because they rely on only one type of input. Multimodal systems can offer broader information, but integrating different data sources introduces challenges involving data synchronization, missing information, computational complexity, and inconsistent data quality. Limited or imbalanced datasets may reduce reliability across different ages, backgrounds, and developmental conditions. Privacy, informed consent, data security, and the risk of false-positive or false-negative predictions are additional concerns. Furthermore, many research models require stronger real-world validation before they can be considered suitable for practical screening. These limitations highlight the need for an accessible, responsible, and carefully validated screening framework.

2.3 Need for the Proposed Project

The proposed NeuroScan AI: Multimodal AI Framework for Early Multi-Disorder Neurodevelopmental Screening and Risk Assessment aims to address the need for accessible preliminary screening, particularly for children in underserved communities. Early identification of possible developmental concerns can help families and caregivers seek appropriate professional assessment sooner. A framework that considers multiple available data modalities may capture a wider range of behavioural indicators than a single-input approach.

CHAPTER 3 REQUIREMENTS AND CONSTRAINTS

3.1 Requirements

3.1.1 Functional Requirements

This section will outline the specific functional expectations of the project. It will describe the core features, logic operations, algorithms, and system behaviours that must be imple-mented to ensure the design achieves its intended purpose. Each requirement will be measur-able, verifiable, and directly linked to the project objectives as mentioned in section 1.3. **This must be based on the chapter 1 objectives and outcomes, existing work as mentioned in chapter 2 [with proper citations] [6].**

Definition: Describe *what the system must do*. These are the **core operations or features** that define the expected behaviour of the design.

Nature:

- Binary: either satisfied or not.
- Tied to correctness and logic.
- Directly linked to the project's *objectives/algorithm*.

Example for a CNTFET-Reversible Multiplier project: The system's core function re-quires it to multiply two 8-bit inputs to produce a corresponding 16-bit output. The design must strictly implement the Urdhva-Tiryagbhyam multiplication sutra and utilize specific reversible logic gates such as the Feynman, Fredkin, Toffoli, and MUF gates. Furthermore, the completed circuit must be synthesizable and fully verifiable using standard simulation tools.

3.1.2 Performance Requirements

This section will define the quantitative expectations for the system's efficiency and robust-ness. It will outline measurable parameters such as speed, power consumption, hardware area, and reliability. These requirements ensure that the design is not only functionally cor-rect but also optimized for practical use under different operating conditions. **This must be based on the chapter 1 objectives and outcomes, existing work as mentioned in chap-ter 2 [with proper citations] [7].**

Definition: Describe *how well the system performs its functions*. These are **quantitative measures or constraints that define efficiency, speed, power, and other performance**

Related Aspect

Nature:

- Measured on a **scale or range** (not just yes/no).
- Often compared against benchmarks..
- Tied to optimization and design quality..

Example for a CNTFET-Reversible Multiplier project: For a CNTFET-Reversible Mul-tiplier project, the system constraints require the **maximum propagation delay** to be less than **X ns**, the **power consumption** to be less than **Y μW**, and the **hardware area** not to ex-ceed **Z logic blocks/gates**. Furthermore, the circuit must **maintain accuracy** across voltage ranges of **A–B V**.The following Table 3.1 lists the summary of requirements of our project **[Citations is com-pulsory for this table]**.

Table 3.1: Summary of Requirements

Req. ID	Description	Category	Target/Expectation
FR1	Implement 8-bit Vedic multiplier using CNTFET and reversible logic [7]	Functional	Functionally correct outputs for all inputs
FR2	Use Urdhva-Tiryagbhyam algorithm [3]	Functional	Parallel multiplication of partial products
FR3	Integrate reversible logic gates (Feynman, Fredkin, Toffoli, MUF) [8]	Functional	Reversible mapping of inputs to outputs
PR1	Reduce power dissipation compared to CMOS	Perfor-mance	$\leq$ 68 mW
PR2	Reduce area utilisation [9]	Perfor-mance	$\leq$ 1750 μm^2 for 8-bit design
PR3	Improve propagation delay [10]	Perfor-mance	$\leq$ 21.6 ns
PR4	Ensure scalability to higher-bit multipliers [11]	Perfor-mance	Design extendable to 16-bit / 32-bit

3.2 Engineering Standards and Codes

[List the **applicable standards, regulations, or codes** that must be followed.]

- **For hardware:** IEEE, ISO, ASTM, IEC, FDA, or safety standards (e.g., ISO 10993

for biocompatibility).

- **For software:** ISO/IEC 25010 (software quality), coding standards, data privacy/ security standards (e.g., HIPAA, GDPR if handling medical data).

Explain briefly how each standard will influence the design choices. Use IEEE citation style where references are available.

Table 3.2: Summary of Standards Considered

Standard	**Domain / Scope**	**Relevance to Project**
IEEE 1364-2005 (Verilog HDL)	Hardware description language	Used for coding and simulating the multiplier architecture
IEEE 1801-2018 (UPF)	Low-power digital design	Provides guidance for analysing low-power intent of the design
IEEE 754-2019	Arithmetic operations	Ensures arithmetic accuracy framework consistency

These standards in Table 3.2 were chosen because they directly relate to the way the circuit was described, simulated, and analysed in this project. Verilog governs the coding and ver-ification process, UPF offers guidance on low-power behaviour, and IEEE 754 reflects the arithmetic accuracy expected in digital systems. Together, they keep the work aligned with accepted practices in circuit design.

3.3 Design Constraints

[Discuss the **realistic constraints** that the project must operate under. These should cover *economic, environmental, sustainability, manufacturability, ethical, health & safety, social, and political* factors. For each, either:]

- Write a short, project-specific 3–5 sentences (if applicable). Be specific about how each affects this project. (or)
- State clearly "Not directly applicable to this project but considered" (so evaluators see you thought about it).

Example:

- **A software-only project** may not have manufacturability constraints → just note "No hardware manufacturing involved; constraint not applicable."
- **A hardware biomedical project** will have strong safety and ethical constraints → must be described in detail.

3.3.1 Economic Constraints

[Discuss cost limits for components, software licenses, prototyping, and testing. Example: "The total hardware budget must not exceed ₹ 25,000."]

- **Software-only:**

  Project must use open-source libraries and student license software to minimize costs. Cloud deployment should fit within free-tier usage limits. No paid datasets will be purchased.
- **Hardware-only:**

  Total component cost must remain under ₹ XX,000. Preference is given to locally available parts to reduce procurement cost. Manufacturing processes (PCB fabrica-tion, 3D printing) must fit within lab resources.
- **Hybrid:**

  Hardware cost is capped at ₹ XX,000 and software tools should rely on open-source frameworks (Python, TensorFlow, Arduino IDE). The budget allocation must cover both hardware procurement and software deployment.

3.3.2 Environmental and Sustainability Constraints

[Explain environmental considerations such as recyclability, power consumption, e-waste, or sustainable materials. Example: "The system should consume < 5W of power to ensure energy efficiency."]

- **Software-only:**

  Cloud computing energy use should be minimized by optimizing algorithms. Data storage must avoid unnecessary redundancy. The software will be designed to run on low-power devices when possible.
- **Hardware-only:**

  Materials selected should be recyclable (e.g.,) aluminium casing, RoHS-compliant components. Device power consumption must remain below X watts to reduce energy use.
- **Hybrid:**

  Software efficiency reduces processing load, lowering device battery drain. Hardware uses low-power ICs and recyclable materials. Together, the system supports sustain-ability by reducing energy and e-waste.

3.3.3 Manufacturability Constraints

[Describe challenges related to fabrication, assembly, or sourcing components. Example: "The PCB must use only components available from local suppliers."]

- **Software-only:**

Software must be compatible with widely available hardware (PCs, Android/iOS de-vices). Deployment should require no specialized or costly platforms.

- **Hardware-only:**
  The design must be realizable with common lab equipment. PCB layout should follow standard manufacturing guidelines. Components should be through-hole or surface-mount types available locally.
- **Hybrid:**
  Software should integrate with microcontrollers (Arduino, Raspberry Pi) that are eas-ily sourced. Hardware must allow repeatable assembly with student-accessible tools.

3.3.4 Ethical and Health & Safety Constraints

[Address issues such as user privacy, ethical responsibility, informed consent, user safety, or bias in algorithms. Example: “Collected data must be anonymized to protect patient identity.”]

Ethical: protecting user privacy, preventing bias in algorithms, obtaining informed consent for data use.

Health and Safety: ensuring the hardware poses no electrical, thermal, or mechanical haz-ards; software avoids harmful recommendations. Explain how relevant standards (e.g., ISO 13485 for medical devices, IEC 62353 for electrical safety, GDPR/HIPAA for data privacy) apply to your project. If risks are minimal, explicitly state this.

- **Software-only:**
  Data privacy must be ensured e.g., anonymization of user inputs, GDPR HIPAA com-pliance if medical data is used. Algorithms must avoid bias against demographic groups.
- **Hardware-only:**
  Device must comply with electrical safety standards (low-voltage operation <12V). Materials contacting users (e.g., biomedical sensors) must be biocompatible. No sharp edges or overheating allowed.
- **Hybrid:**
  Hardware must meet electrical and physical safety requirements, while software en-sures secure data handling and unbiased predictions. Together, the system must protect both user safety and ethical responsibility.

3.3.5 Social Constraints

[Explain how cultural, accessibility, or usability factors shape the design. Example: “The interface should support multiple languages for inclusivity.”]

- **Software-only:**
  User interface must be accessible (multilingual support, font sizes for visually im-paired). System should avoid stigmatization (neutral wording in reports).
- **Hardware-only:**
  Device form factor must be comfortable and socially acceptable (wearable design, lightweight). Design should avoid invasive procedures.
- **Hybrid:**
  Combined system must be user-friendly and culturally sensitive. Hardware must be comfortable, while software output must be easy to interpret and not stigmatizing.

3.3.6 Political/Regulatory Constraints

[Note any government regulations, approvals, or policies affecting deployment. Example: "Deployment in healthcare settings must comply with national medical device regulations."]

- **Software-only:**
  Compliance with data protection regulations (GDPR, HIPAA). No use of blacklisted or non-licensed APIs. Deployment must respect local internet/data laws.
- **Hardware-only:**
  If intended for healthcare, the device must follow local medical device regulations. Export-controlled components (if any) must not be used.
- **Hybrid:**
  Hardware follows safety/medical regulations; software follows privacy/data laws. De-ployment in a healthcare or public environment requires meeting both sets of compli-ance requirements.

3.4 Data Ethics and Management

Only for Projects which use a dataset.

Data Collection: All datasets used in this project were obtained from [public reposito-ries/surveys/experiments] with appropriate permissions. Any personal/health data collection followed informed consent procedures.

Data Privacy: Sensitive data was anonymized before processing. Identifiable attributes (e.g., names, IDs, health markers) were removed or masked.

Data Storage and Security: Project data was stored in [encrypted/local/secured cloud] sys-tems with access restricted to team members.

Compliance: The project followed applicable standards such as GDPR, HIPAA, IEEE Code

of Ethics for handling sensitive data.

Data Sharing: Processed datasets, source code, and results are shared in the appendices/ GitHub repository, ensuring reproducibility and transparency.

3.1 Requirements

The NeuroScan AI project aims to develop a multimodal artificial intelligence framework for early screening of multiple neurodevelopmental and neuromental disorders in children. The system focuses on processing relevant input data, identifying potential patterns, and generating preliminary risk assessment results. The requirements are divided into functional and performance requirements.

3.1.1 Functional Requirements

The system shall accept suitable input data, preprocess the data, and extract relevant features for analysis. It shall support the integration of multiple modalities, such as facial or behavioural video features, speech or audio features, and other relevant assessment information, depending on dataset availability. The AI model shall analyse the extracted features and generate a preliminary screening result. The system shall provide an accessible interface for users to submit inputs and view results. It shall also handle invalid or incomplete inputs and protect sensitive information throughout processing.

3.1.2 Performance Requirements

The framework should provide consistent results, reasonable processing time, and reliable operation. Model performance should be evaluated using accuracy, precision, recall, F1-score, sensitivity, and specificity, where appropriate. Testing should assess whether the model performs consistently across different data samples and relevant groups. No accuracy target should be claimed until it has been established through actual testing. The system should also be designed to support future expansion to additional disorders and data modalities.

3.2 Engineering Standards and Codes

The project should follow relevant software engineering, data protection, and ethical principles. ISO/IEC 25010 can guide software quality assessment, while applicable data protection requirements should guide the collection, storage, processing, and sharing of sensitive information. The IEEE Code of Ethics provides guidance on responsible engineering practice. If the framework is used in a clinical setting, applicable healthcare regulations and

medical-device requirements must be reviewed before deployment. These considerations support software reliability, privacy, transparency, and responsible use.

3.3 Design Constraints

3.3.1 Economic Constraints

The project should prioritize open-source software libraries and publicly available datasets with suitable usage permissions. Development should remain within student resources and available computing facilities. Paid services and specialized hardware should be avoided unless necessary.

3.3.2 Environmental and Sustainability Constraints

As the framework is primarily software-based, environmental impact mainly relates to computing power, storage, and energy consumption. Efficient preprocessing and model selection can reduce unnecessary computational requirements. Reusing available computing resources can also help minimize additional electronic waste.

3.3.3 Manufacturability Constraints

The project does not require physical product manufacturing. Its main constraints concern software compatibility, computing resources, dataset availability, and integration of different data-processing modules. The framework should use widely available development tools wherever possible.

3.3.4 Ethical and Health and Safety Constraints

Children's behavioural and health-related data are sensitive and must be handled responsibly. Data collection and use must follow applicable consent, permission, and privacy requirements. The model must not be presented as providing a confirmed medical diagnosis. Its output should be treated as preliminary screening information, and suspected developmental concerns should be referred to qualified professionals. The system should also consider bias and avoid misleading or stigmatizing results.

3.3.5 Social Constraints

The system should be simple to use and suitable for caregivers and other intended users with different levels of technical knowledge. Clear language, accessible presentation, and affordability are important, especially for underserved communities. The output should communicate limitations without creating unnecessary fear or stigma.

3.3.6 Political and Regulatory Constraints

Any deployment involving children's health information must comply with applicable data protection, consent, and healthcare requirements. The project should use datasets and software under their permitted licenses. Clinical or public deployment should not be claimed until the necessary regulatory and institutional requirements have been assessed.